



★ Excel :-

- (*) Introduction.
- (*) Basic Functions / Text Functions.
- (*) Math Functions.
- (*) Date & Time Functions.
- (*) Financial Functions.
- (*) Sorting and Filtering Functions.
- (*) Data Validation.
- (*) Text to Columns.
- (*) Conditional Formatting & Conditional Functions.
- (*) Lookups
- (*) Pivot Tables.
- (*) Power Pivot Tables.
- (*) Power Query - PowerBI tool for data transferring.
- (*) Data Visualization.
- (*) AI with Excel.
- (*) Dashboard Creation & Report Generation.
- (*) Vector - One Column.
- (*) Matrix - Rows & Columns.
- (*) Structured - Tabular Format.

★ Meaning of Microsoft Excel :- A popular spreadsheet program developed by Microsoft that lets you organize, calculate & analyze data in rows and columns.

Its Key features are Grid System, formulas & functions, Charts & Graphs, Data Tools.

It is most commonly used for tracking personal or business money i.e., budgeting, for storing lists names and for reporting data in a neat tables.



★ Worksheet, Rows, Columns and Cells :-

The fundamental structure of Excel. A workbook is made of worksheets which consists of horizontal rows, vertical columns and individual cells where data is typed.

★ What's the need of Excel :-

Excel is essential for efficient data management, financial tracking, and analysis, handling various data types such as text, numbers, dates and boolean values.

★ Types of data in Excel :-

1. Text (string) $\Rightarrow$ Letters, words or alphanumeric combinations (like names or addresses).
2. Numbers (numeric) $\Rightarrow$ Raw digits, decimals or integers used for calculations.
3. Boolean $\Rightarrow$ Logical values that result in either TRUE or FALSE.
4. Dates and Time $\Rightarrow$ Specific calendar dates or time stamps formatted so Excel can calculate durations or time differences.

★ Important Excel Features :-

1. Data Cleaning $\Rightarrow$ The process of detecting and correcting corrupt, inaccurate or duplicate records from a dataset.
2. Formulas and Functions $\Rightarrow$ Built-in tools and mathematical expressions used to perform calculations, automate tasks & manipulate data.



3. Data Visualization $\Rightarrow$ Turning numbers and tables into charts or graphs to easily spot trends & insights.
4. Pivot Table $\Rightarrow$ An advanced tool used to quickly summarize, group, and analyze large sets of data without writing complex formulas.
5. Sorting and Filtering $\Rightarrow$ Techniques to organize data in a specific sequence (alphabetical, numerical) or temporarily hide rows that don't match specific criteria.
6. Data Validation $\Rightarrow$ A rule-setting feature that restricts what users can type into a specific cell to prevent data entry errors.

★ Advantages and Disadvantages of Excel :-

- (*) Advantages $\Rightarrow$ Highly accessible, versatile for tasks ranging from simple lists to complex financial modeling & widely used across industries.
- (*) Disadvantages $\Rightarrow$ Can slow down or crash massive datasets (millions of rows), lacks built-in multi-user real-time editing compared to cloud datasets, and is prone to human error in complex formula building.



★ Excel Text Formulas :-

(*) Case Conversion Formulas :

1. UPPER $\Rightarrow$ Converts all letters in a text string to uppercase.

Syntax : =UPPER (text)

Example : =UPPER ("Rochitha")

Output : ROCHITHA.

2. LOWER $\Rightarrow$ Converts all letters in a text string to lowercase.

Syntax : =LOWER (text)

Example : =LOWER ("Rochitha")

Output : rochitha.

3. PROPER $\Rightarrow$ Capitalizes the first letter of each word in a text string and converts all other letters to lowercase.

Syntax : =PROPER (text)

Example : =PROPER ("Rochitha Shetty")

Output : Rochitha Shetty.

(*) Text Length and Extraction Formulas :

1. LEN $\Rightarrow$ Returns the total number of characters in a text string (including spaces and punctuation).

Syntax : =LEN (text)

Example : =LEN ("Data Analyst").

Output : 12.



2. LEFT $\Rightarrow$ Extracts a specified number of characters from the beginning (left side) of a text string.

Syntax: =LEFT(text, [num-chars])

(If num-chars is omitted, it defaults to 1).

Example: =LEFT("Rochitha", 4)

Output: Roch.

3. RIGHT $\Rightarrow$ Extracts a specified number of characters from the end (right side) of a text string but not reverse.

Syntax: =RIGHT(text, [num-chars]).

Example: =RIGHT("Rochitha", 4)

o/p: itha

=RIGHT("Rochitha")

o/p: a.

4. MID $\Rightarrow$ Extracts a specific number of characters from the middle of a text string, by giving a starting position and length.

Syntax: =MID(text, start-num, num-chars).

Example: =MID("Rochitha", 4, 5)

output: hitha.

(*) Dividing Text into First and Second Halves:

To split a text string in cell A1 exactly in half, you can combine LEN, LEFT, RIGHT and ROUND, ROUNDUP & ROUNDDOWN (This is especially when a name has an odd number of characters like 5 letters). Dividing 5 by 2 gives 2.5, and rounding functions let you control whether the extra character goes to the first half



or the second half.

(*) Example : Splitting an Odd-length Name
(Rochi = 5 letters)

1. First Using RoundUp :-

> First Half Formula :

Syntax : $=\text{LEFT}(A1, \text{ROUNDUP}(\text{LEN}(A1)/2, 0))$

Output : Extracts 3 letters from the left
left. $\rightarrow$ Roc

2. > Second Half Formula :

Syntax : $=\text{RIGHT}(A1, \text{ROUNDUP}(\text{LEN}(A1)/2, 0))$

Output : Extracts the remaining 2 letters
 $\rightarrow$ hi

2. Using ROUNDDOWN :-

> First Half Formula $\Rightarrow$

Syntax : $=\text{LEFT}$

Syntax : $=\text{LEFT}(A1, \text{ROUNDDOWN}(\text{LEN}(A1)/2, 0))$

Output : Extracts 2 letters from the left.
 $\rightarrow$ Ro

> Second Half Formula $\Rightarrow$

Syntax : $=\text{RIGHT}(A1, \text{ROUNDDOWN}(\text{LEN}(A1)/2, 0))$

Output : Extracts the remaining 3 letters.
 $\rightarrow$ chi.

3. Using Standard ROUND :- Standard ROUND

$\Rightarrow$ It behaves the same as ROUNDUP here,
because 2.5 rounds up in standard
mathematics.



> First Half Formula:

Syntax: =LEFT(A1, ROUND(LEN(A1)/2, 0))

Output: Roc

> Second Half formula:

Syntax: =RIGHT(A1, ROUND(LEN(A1)/2, 0))

Output: hi

(*) Rounding Formulas:

1. ROUND $\Rightarrow$ Rounds a number to a specified number of digits based on standard mathematical rounding rules (if the next digit is 5 or greater, it rounds up otherwise it rounds down).

Syntax: =ROUND(number, num-digits)

Example: =ROUND(3.141596, 2)

Output: 3.14

2. ROUNDUP $\Rightarrow$ Always rounds a number up (away from zero) to a specified number of digits, regardless of the following digit.

Syntax: =ROUNDUP(number, num-digits)

Example: =ROUNDUP(3.141296, 2)

Output: 3.15

3. ROUNDDOWN $\Rightarrow$ Always rounds a number down (toward zero) to a specified number of digits, effectively truncating extra decimal places without standard rounding.

Syntax: =ROUNDDOWN(number, num-digits)

Example: =ROUNDDOWN(3.14999, 2)

Output: 3.14